

# AI-Led Hackathon

## Resource & Tooling Guide

*Build AI Agents Powered by Telecom Network Intelligence*

Companion reference for GSMA MENA — Regional Edition

### 1. Overview

This guide expands on the proposed hackathon direction: moving participants beyond simply consuming CAMARA APIs and toward building AI-powered solutions that use those APIs as trusted, real-time data sources. The deliverables expected are AI agents, copilots, and autonomous workflows that draw on telecom signals such as Number Verification, SIM Swap, Device Status, Location Verification, and Network Quality to make intelligent decisions.

Every tool listed below has a Free plan or a Freemium tier that is sufficient to build, demo, and submit a working hackathon project — no credit card or paid commitment required to get started.

#### Suggested Focus Areas

- AI for Fraud Detection and Digital Identity
- AI-powered Financial Services
- AI Healthcare Assistants
- Smart Mobility and Logistics Agents
- Customer Support and Operations Copilots
- Telecom-specific AI Agents

### 2. AI Agent Building Platforms

These platforms let participants design agentic workflows visually or in code. The first four are low-barrier — ideal for students and first-time AI builders. The rest scale up to multi-agent and production-grade systems.

#### Low-code / Visual builders (great for beginners)

| Tool    | What it does                                                                                           | Tier | Link                  |
|---------|--------------------------------------------------------------------------------------------------------|------|-----------------------|
| Flowise | Open-source drag-and-drop builder for LLM apps and agents. Self-host free or use the Cloud free trial. | Free | <a href="#">Visit</a> |

| Tool            | What it does                                                                                              | Tier     | Link                  |
|-----------------|-----------------------------------------------------------------------------------------------------------|----------|-----------------------|
| <b>Langflow</b> | Visual framework for building multi-agent and RAG apps; open-source, integrates with most LLMs and tools. | Free     | <a href="#">Visit</a> |
| <b>n8n</b>      | Workflow automation with native AI nodes; community edition is free and self-hostable.                    | Freemium | <a href="#">Visit</a> |
| <b>Dify</b>     | Open-source LLMOps platform — build chatbots, agents and RAG apps without writing much code.              | Freemium | <a href="#">Visit</a> |
| <b>Make</b>     | No-code automation with AI modules; free tier covers up to 1,000 operations per month.                    | Freemium | <a href="#">Visit</a> |

### Code-first agent frameworks (for advanced participants)

| Tool                     | What it does                                                                                               | Tier     | Link                  |
|--------------------------|------------------------------------------------------------------------------------------------------------|----------|-----------------------|
| <b>CrewAI</b>            | Python framework for multi-agent systems where agents collaborate with defined roles and tools.            | Free     | <a href="#">Visit</a> |
| <b>LangGraph</b>         | Graph-based orchestration for stateful, long-running agents — built on LangChain.                          | Free     | <a href="#">Visit</a> |
| <b>OpenAI Agents SDK</b> | Lightweight SDK to build agents that can autonomously call CAMARA APIs as tools. Pay only for tokens used. | Freemium | <a href="#">Visit</a> |
| <b>Microsoft AutoGen</b> | Multi-agent conversation framework from Microsoft Research; open-source.                                   | Free     | <a href="#">Visit</a> |
| <b>Pydantic AI</b>       | Type-safe Python agent framework focused on production reliability.                                        | Free     | <a href="#">Visit</a> |
| <b>LlamaIndex</b>        | Data framework and agent toolkit, strong for RAG over private and API data.                                | Freemium | <a href="#">Visit</a> |
| <b>Smolagents (HF)</b>   | Hugging Face's minimal agent library — agents that write Python to call tools.                             | Free     | <a href="#">Visit</a> |

### Enterprise-grade options

| Tool                                  | What it does                                                                | Tier     | Link                  |
|---------------------------------------|-----------------------------------------------------------------------------|----------|-----------------------|
| <b>Google Vertex AI Agent Builder</b> | Production-ready agents on GCP; new users get \$300 in free credits.        | Freemium | <a href="#">Visit</a> |
| <b>AWS Bedrock Agents</b>             | Managed agent service on AWS; free tier credits available for new accounts. | Freemium | <a href="#">Visit</a> |

| Tool                    | What it does                                                                                    | Tier     | Link                  |
|-------------------------|-------------------------------------------------------------------------------------------------|----------|-----------------------|
| <b>Azure AI Foundry</b> | Microsoft's end-to-end agent and model platform; free Azure credits for students and new users. | Freemium | <a href="#">Visit</a> |

### 3. LLMs and Model APIs (Free Access)

Agents need a brain. These providers offer either a generous free tier, free API credits, or fully open models you can self-host.

#### Hosted APIs with a free tier

| Tool                              | What it does                                                                                         | Tier     | Link                  |
|-----------------------------------|------------------------------------------------------------------------------------------------------|----------|-----------------------|
| <b>Google AI Studio (Gemini)</b>  | Free API key for Gemini models including Gemini 2.5 Flash and Pro; very generous request limits.     | Free     | <a href="#">Visit</a> |
| <b>Groq</b>                       | Extremely fast inference on open models (Llama, Mixtral, Qwen). Free API tier with high rate limits. | Freemium | <a href="#">Visit</a> |
| <b>OpenRouter</b>                 | One API for many models — includes free-tier routing to several open models.                         | Freemium | <a href="#">Visit</a> |
| <b>Cohere</b>                     | Trial API keys for Command and embedding models; great for RAG.                                      | Freemium | <a href="#">Visit</a> |
| <b>Mistral La Plateforme</b>      | Free experimentation tier on Mistral's hosted API.                                                   | Freemium | <a href="#">Visit</a> |
| <b>Together AI</b>                | Hosted open-source models with free starter credits.                                                 | Freemium | <a href="#">Visit</a> |
| <b>Hugging Face Inference API</b> | Run thousands of open models via API; free serverless tier.                                          | Freemium | <a href="#">Visit</a> |
| <b>Cerebras Inference</b>         | Ultra-fast Llama inference with a free developer tier.                                               | Freemium | <a href="#">Visit</a> |

#### Run models locally (zero cost, full privacy)

| Tool             | What it does                                                                                  | Tier | Link                  |
|------------------|-----------------------------------------------------------------------------------------------|------|-----------------------|
| <b>Ollama</b>    | One-line install to run Llama, Mistral, Qwen, Gemma and others on your own laptop.            | Free | <a href="#">Visit</a> |
| <b>LM Studio</b> | Desktop app to download and chat with local open-source LLMs; OpenAI-compatible local server. | Free | <a href="#">Visit</a> |

| Tool             | What it does                                                            | Tier | Link                  |
|------------------|-------------------------------------------------------------------------|------|-----------------------|
| <b>vLLM</b>      | High-throughput inference engine for self-hosting open models on a GPU. | Free | <a href="#">Visit</a> |
| <b>llama.cpp</b> | Run quantized LLMs efficiently on CPU and consumer GPUs.                | Free | <a href="#">Visit</a> |

## 4. Vector Databases and Agent Memory

For retrieval-augmented agents — letting your agent remember conversations, ground answers in documents, or search telecom logs semantically.

| Tool                         | What it does                                                                     | Tier     | Link                  |
|------------------------------|----------------------------------------------------------------------------------|----------|-----------------------|
| <b>Chroma</b>                | Open-source embedding database — runs locally with one pip install.              | Free     | <a href="#">Visit</a> |
| <b>Qdrant</b>                | Open-source vector search engine; managed cloud has a 1 GB free cluster.         | Freemium | <a href="#">Visit</a> |
| <b>Weaviate</b>              | Vector DB with built-in hybrid search; free sandbox cluster available.           | Freemium | <a href="#">Visit</a> |
| <b>Pinecone</b>              | Managed vector DB; Starter tier is free forever with one index.                  | Freemium | <a href="#">Visit</a> |
| <b>Milvus / Zilliz Cloud</b> | Open-source vector DB; Zilliz offers a free serverless tier.                     | Freemium | <a href="#">Visit</a> |
| <b>Supabase pgvector</b>     | Postgres + pgvector on Supabase free tier — vectors plus a real database in one. | Freemium | <a href="#">Visit</a> |
| <b>Mem0</b>                  | Drop-in memory layer for agents; open-source plus a hosted free tier.            | Freemium | <a href="#">Visit</a> |

## 5. CAMARA APIs — The Trusted Data Layer

CAMARA is the open-source project (under the Linux Foundation, GSMA-aligned) that standardises telecom network APIs. These are the signals your AI agents will consume.

| Tool                  | What it does                                                     | Tier | Link                  |
|-----------------------|------------------------------------------------------------------|------|-----------------------|
| <b>CAMARA Project</b> | Official project page — specifications, governance, and roadmap. | Free | <a href="#">Visit</a> |
| <b>CAMARA GitHub</b>  | All API definitions, OpenAPI specs and sandbox info.             | Free | <a href="#">Visit</a> |

| Tool                         | What it does                                                  | Tier     | Link                  |
|------------------------------|---------------------------------------------------------------|----------|-----------------------|
| <b>GSMA Open Gateway</b>     | The commercial framework operators use to expose CAMARA APIs. | Free     | <a href="#">Visit</a> |
| <b>Nokia Network as Code</b> | Free developer portal with CAMARA-aligned APIs and sandbox.   | Freemium | <a href="#">Visit</a> |

## 6. Development and Deployment (Free Tiers)

### Coding and AI pair-programming

| Tool                      | What it does                                                                             | Tier     | Link                  |
|---------------------------|------------------------------------------------------------------------------------------|----------|-----------------------|
| <b>Claude (Anthropic)</b> | Conversational AI assistant for coding, design and writing — free plan available.        | Freemium | <a href="#">Visit</a> |
| <b>ChatGPT</b>            | OpenAI's assistant with a capable free tier.                                             | Freemium | <a href="#">Visit</a> |
| <b>Google Gemini</b>      | Google's assistant with a strong free tier.                                              | Freemium | <a href="#">Visit</a> |
| <b>GitHub Copilot</b>     | AI pair programmer in your IDE — free for students and verified open-source maintainers. | Freemium | <a href="#">Visit</a> |
| <b>Cursor</b>             | AI-first code editor with a free Hobby plan.                                             | Freemium | <a href="#">Visit</a> |
| <b>Windsurf</b>           | Agentic IDE from Codeium with a free tier.                                               | Freemium | <a href="#">Visit</a> |
| <b>Continue.dev</b>       | Open-source AI coding assistant for VS Code and JetBrains.                               | Free     | <a href="#">Visit</a> |
| <b>Bolt.new</b>           | Browser-based AI full-stack builder; free daily tokens.                                  | Freemium | <a href="#">Visit</a> |
| <b>v0 by Vercel</b>       | Generate React UI from prompts; free tier monthly credits.                               | Freemium | <a href="#">Visit</a> |
| <b>Lovable</b>            | AI app builder for full-stack web apps; free starter credits.                            | Freemium | <a href="#">Visit</a> |

### Hosting and deployment

| Tool           | What it does                                                  | Tier     | Link                  |
|----------------|---------------------------------------------------------------|----------|-----------------------|
| <b>Vercel</b>  | Deploy Next.js / web frontends globally; generous Hobby tier. | Freemium | <a href="#">Visit</a> |
| <b>Netlify</b> | Static and serverless hosting with a free Starter plan.       | Freemium | <a href="#">Visit</a> |
| <b>Render</b>  | Free web services, static sites and PostgreSQL.               | Freemium | <a href="#">Visit</a> |

| Tool                             | What it does                                                           | Tier     | Link                  |
|----------------------------------|------------------------------------------------------------------------|----------|-----------------------|
| <b>Railway</b>                   | Deploy backends and databases; \$5 free trial credits.                 | Freemium | <a href="#">Visit</a> |
| <b>Hugging Face Spaces</b>       | Free hosting for Gradio / Streamlit demos — ideal for hackathon demos. | Freemium | <a href="#">Visit</a> |
| <b>Streamlit Community Cloud</b> | One-click deployment for Python data and AI apps.                      | Free     | <a href="#">Visit</a> |
| <b>Cloudflare Workers</b>        | Edge functions and Workers AI; generous free plan.                     | Freemium | <a href="#">Visit</a> |
| <b>Replit</b>                    | Cloud IDE with AI features and free hosting for small projects.        | Freemium | <a href="#">Visit</a> |

#### Databases and backend

| Tool                 | What it does                                                                    | Tier     | Link                  |
|----------------------|---------------------------------------------------------------------------------|----------|-----------------------|
| <b>Supabase</b>      | Postgres + auth + storage + realtime; free tier covers most hackathon projects. | Freemium | <a href="#">Visit</a> |
| <b>Firebase</b>      | Google's backend with Spark free plan.                                          | Freemium | <a href="#">Visit</a> |
| <b>Neon</b>          | Serverless Postgres with a free tier.                                           | Freemium | <a href="#">Visit</a> |
| <b>MongoDB Atlas</b> | Free 512 MB managed MongoDB cluster.                                            | Freemium | <a href="#">Visit</a> |
| <b>Upstash</b>       | Serverless Redis and Kafka with a generous free tier.                           | Freemium | <a href="#">Visit</a> |

## 7. Voice, Vision and Speech (Useful for Telecom Copilots)

| Tool                  | What it does                                                                           | Tier     | Link                  |
|-----------------------|----------------------------------------------------------------------------------------|----------|-----------------------|
| <b>ElevenLabs</b>     | Realistic text-to-speech and voice agents; free tier with monthly character allowance. | Freemium | <a href="#">Visit</a> |
| <b>Deepgram</b>       | Speech-to-text API with \$200 free credits for new accounts.                           | Freemium | <a href="#">Visit</a> |
| <b>AssemblyAI</b>     | Speech AI with free hours and strong multilingual support.                             | Freemium | <a href="#">Visit</a> |
| <b>OpenAI Whisper</b> | Open-source speech-to-text model you can run yourself for free.                        | Free     | <a href="#">Visit</a> |
| <b>Vapi</b>           | Build AI phone agents — handy for telecom-themed bots; free credits.                   | Freemium | <a href="#">Visit</a> |

| Tool                  | What it does                                                          | Tier     | Link                  |
|-----------------------|-----------------------------------------------------------------------|----------|-----------------------|
| <b>LiveKit Agents</b> | Open-source realtime voice and video agents framework.                | Freemium | <a href="#">Visit</a> |
| <b>Twilio</b>         | Programmable voice, SMS and WhatsApp; trial credits for new accounts. | Freemium | <a href="#">Visit</a> |

## 8. Learning Resources (All Free)

| Tool                                     | What it does                                                          | Tier     | Link                  |
|------------------------------------------|-----------------------------------------------------------------------|----------|-----------------------|
| <b>DeepLearning.AI Short Courses</b>     | Free 1-hour courses on LangChain, CrewAI, LangGraph, agents, and RAG. | Free     | <a href="#">Visit</a> |
| <b>Hugging Face Agents Course</b>        | Free end-to-end course on building AI agents.                         | Free     | <a href="#">Visit</a> |
| <b>LangChain Academy</b>                 | Free courses on LangChain and LangGraph.                              | Free     | <a href="#">Visit</a> |
| <b>Microsoft AI Agents for Beginners</b> | Free 10-lesson curriculum on GitHub.                                  | Free     | <a href="#">Visit</a> |
| <b>Google Cloud Skills Boost</b>         | Free labs on Vertex AI and Gemini.                                    | Freemium | <a href="#">Visit</a> |
| <b>Anthropic Courses</b>                 | Free Claude prompting and tool-use courses on GitHub.                 | Free     | <a href="#">Visit</a> |
| <b>OpenAI Cookbook</b>                   | Free recipes and examples covering agents, RAG and function calling.  | Free     | <a href="#">Visit</a> |
| <b>Prompt Engineering Guide</b>          | Comprehensive free guide and reference.                               | Free     | <a href="#">Visit</a> |
| <b>Full Stack Deep Learning</b>          | Free lectures on building LLM-powered apps end to end.                | Free     | <a href="#">Visit</a> |

## 9. Productivity AI (For Demos, Slides and Pitches)

| Tool            | What it does                                               | Tier     | Link                  |
|-----------------|------------------------------------------------------------|----------|-----------------------|
| <b>Gamma</b>    | AI-generated presentations and docs; free starter credits. | Freemium | <a href="#">Visit</a> |
| <b>Canva AI</b> | Design tool with Magic Studio AI features; free plan.      | Freemium | <a href="#">Visit</a> |

| Tool                   | What it does                                                     | Tier     | Link                  |
|------------------------|------------------------------------------------------------------|----------|-----------------------|
| <b>Tome</b>            | AI storytelling and pitch decks.                                 | Freemium | <a href="#">Visit</a> |
| <b>Beautiful.ai</b>    | AI-assisted slide design with a free trial.                      | Freemium | <a href="#">Visit</a> |
| <b>Excalidraw + AI</b> | Free whiteboard with an AI diagram generator.                    | Free     | <a href="#">Visit</a> |
| <b>Eraser AI</b>       | Generates architecture diagrams from prompts.                    | Freemium | <a href="#">Visit</a> |
| <b>Napkin AI</b>       | Turns text into visuals automatically — useful for pitch slides. | Freemium | <a href="#">Visit</a> |
| <b>NotebookLM</b>      | Google's free research notebook with audio overviews.            | Free     | <a href="#">Visit</a> |
| <b>Perplexity</b>      | AI research assistant with a strong free tier.                   | Freemium | <a href="#">Visit</a> |

## 10. Suggested Starter Architectures

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Three reference combinations participants can pick on day one — all components below have a free tier.

### Beginner stack (no-code first)

- UI: Streamlit Community Cloud
- Agent: Flowise or Langflow (visual)
- LLM: Google AI Studio (Gemini Flash, free)
- Memory: Chroma (local)
- Data source: CAMARA API sandbox (Nokia or Vonage)

### Intermediate stack (Python)

- UI: Next.js on Vercel
- Agent: CrewAI or LangGraph
- LLM: Groq (Llama 3.3 70B, free tier)
- Memory: Supabase pgvector
- Voice: Deepgram + ElevenLabs (free credits)
- Data source: CAMARA APIs exposed as agent tools

### Advanced multi-agent stack

- Orchestration: LangGraph or AutoGen
- Specialist agents: CrewAI roles per focus area
- LLM mix: Gemini 2.5 Pro for reasoning, Groq for fast actions
- Memory: Qdrant Cloud free cluster + Mem0
- Deployment: Hugging Face Space or Cloudflare Workers
- Telecom signals: CAMARA APIs through Aduna or operator sandbox



## 11. Tips for Participants

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- Pick one focus area on day one and resist scope creep — judges prefer one polished agent over five half-built ones.
- Treat each CAMARA API as a tool the agent decides when to call, not a button the user presses. That is what makes a solution 'agentic'.
- Have a clear fallback when an API or model is rate-limited — agents that gracefully degrade demo much better.
- Cache demo data. Live API calls fail at the worst moment; a recorded fallback keeps the demo running.
- Show the agent's reasoning trace on screen during the demo — judges love seeing the 'thinking'.
- Pick free tiers that don't require a credit card so the whole team can sign up quickly.
- Read the rate limits before the hackathon — Groq, Gemini and Cerebras free tiers are generous but not infinite.